



StockPilot

Inventory Management System

Part 1 of 2 — **Product & Features Walkthrough**

Every screen, KPI and the AI reorder advisor — explained for interview prep.

Next.js 16

React 19

PostgreSQL + Prisma

NextAuth

Recharts

Claude / Gemini AI

Prepared 14 July 2026 · Author: Bhanu Prakash

o • What StockPilot is

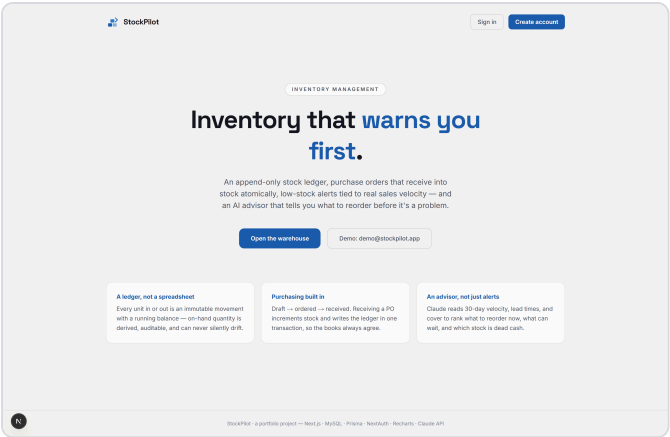
A full-stack inventory management console built on one principle: **stock is a ledger, not a number**. On-hand quantity is never edited directly — it is derived from an append-only stock-movement ledger. Purchase orders receive into stock atomically, low-stock alerts are tied to real 30-day sales velocity, and an AI advisor recommends what to reorder before a stockout.

Interviewer one-liner: "It's a Next.js 16 app-router project where server components read PostgreSQL through Prisma; every stock write is a row-locked transaction that appends to an immutable ledger and rejects oversell with a 409; and an AI reorder advisor uses Claude tool-use (with a Gemini and heuristic fallback) to rank SKUs by stockout risk."

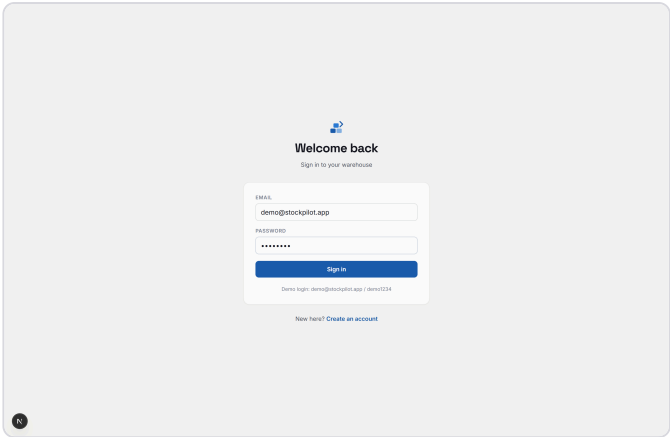
LAYER	CHOICE
Framework / UI	Next.js 16 (App Router, server components) · React 19 · TypeScript · Tailwind 4
Data	PostgreSQL + Prisma ORM (singleton client)
Auth	NextAuth (Credentials, JWT sessions) · bcrypt password hashing
Charts	Recharts (CVD-validated palette)
AI advisor	Anthropic Claude (tool-use) → Google Gemini (REST) → deterministic heuristic

1 • Authentication

Email/password login via NextAuth with JWT sessions; passwords are bcrypt-hashed. The login page pre-fills the demo operator; every protected page redirects to `/login` when there is no session.



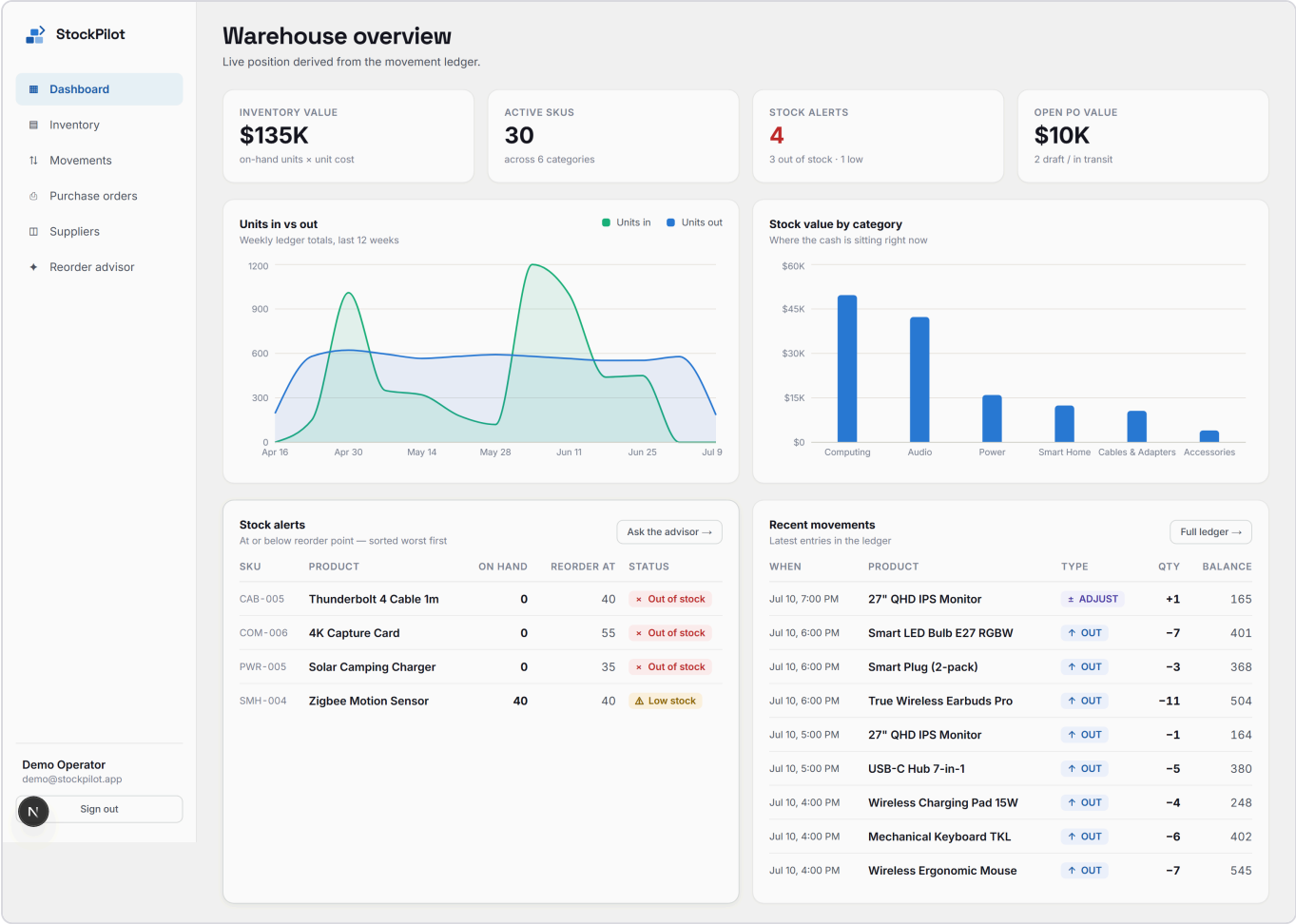
Marketing landing page (redirects to the dashboard if already signed in)



Login — demo@stockpilot.app / demo1234

2 · Dashboard — "Warehouse overview"

The live position, derived entirely from the movement ledger.



Dashboard: four KPI tiles, the in/out flow chart, stock value by category, plus the worst-first alert table and the recent-movements ledger feed

The four KPI tiles

TILE	VALUE	MEANING
Inventory value	\$135K	On-hand units × unit cost across all SKUs
Active SKUs	30	Distinct products, across 6 categories
Stock alerts	4	At/below reorder point — 3 out of stock, 1 low (warn tone)
Open PO value	\$10K	Value of draft / in-transit purchase orders

The panels

- **Units in vs out** — weekly ledger totals for the last 12 weeks; green = units received (IN), blue = units shipped (OUT). The spikes are receiving events against steady outflow.
- **Stock value by category** — where cash is currently tied up (Computing and Audio dominate).
- **Stock alerts** — SKUs at or below their reorder point, sorted worst-first, with an "Ask the advisor →" shortcut.

- **Recent movements** — the latest ledger entries: each row shows type (IN / OUT / $\pm$ ADJUST), quantity, and the resulting running balance snapshot.

3 • Inventory, Movements, Purchase Orders & Suppliers

StockPilot

Dashboard

Inventory

Movements

Purchase orders

Suppliers

Recorder advisor

Demo Operator
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Sign out

Inventory

30 SKUs - stock changes only through the ledger, never by editing a row.

Search SKU, name, supplier...

AB

All

In stock

Low stock

Out of stock

30 shown - 150K on hand

SKU	PRODUCT	CATEGORY	SUPPLIER	COST	PRICE	ON HAND	SELLSDAY	STATUS	
ACC-001	Adjustable Phone Tripod	Accessories	Baltic Electronics OÜ	\$7.00	\$24.00	199	1.83	In stock	688 Delete
ACC-002	Laptop Sleeve 14" Felt	Accessories	Baltic Electronics OÜ	\$5.00	\$22.00	182	1.3	In stock	688 Delete
ACC-003	Cable Management Kit	Accessories	Baltic Electronics OÜ	\$3.50	\$14.00	274	2.7	In stock	688 Delete
ACC-004	Screen Cleaning Kit Pro	Accessories	Baltic Electronics OÜ	\$2.50	\$8.00	250	2.73	In stock	688 Delete
AUD-001	Studio Monitor Headphones	Audio	Kaizen Audio Works	\$48.00	\$119.00	341	2.5	In stock	688 Delete
AUD-002	True Wireless Earbuds Pro	Audio	Kaizen Audio Works	\$16.00	\$78.00	504	6	In stock	688 Delete
AUD-003	USB Condenser Microphone	Audio	Kaizen Audio Works	\$31.00	\$89.00	192	1.5	In stock	688 Delete
AUD-004	Bluetooth Party Speaker XL	Audio	Kaizen Audio Works	\$62.00	\$159.00	34	1.07	In stock	688 Delete
AUD-005	Vinyl Turntable Classic	Audio	Kaizen Audio Works	\$74.00	\$189.00	64	0	In stock	688 Delete
CAB-001	USB-C to HDMI Cable 2m	Cables & Adapters	Rhine Cable GmbH	\$4.20	\$15.00	728	7.23	In stock	688 Delete
CAB-002	USB-C Hub 7-in-1	Cables & Adapters	Rhine Cable GmbH	\$13.00	\$42.00	380	3.6	In stock	688 Delete
CAB-003	Braided Lightning Cable 1m	Cables & Adapters	Rhine Cable GmbH	\$2.80	\$12.00	607	4.67	In stock	688 Delete
CAB-004	Ethernet Cat-6 Cable 5m	Cables & Adapters	Rhine Cable GmbH	\$3.10	\$11.00	291	3.17	In stock	688 Delete
CAB-005	Thunderbolt 4 Cable 1m	Cables & Adapters	Rhine Cable GmbH	\$15.00	\$45.00	0	0	Out of stock	688 Delete
COM-001	Mechanical Keyboard TKL	Computing	Nordic Peripherals AB	\$38.00	\$99.00	402	3.73	In stock	688 Delete
COM-002	Wireless Ergonomic Mouse	Computing	Nordic Peripherals AB	\$14.00	\$39.00	545	4.77	In stock	688 Delete
COM-003	Y885 Streaming Webcam	Computing	Nordic Peripherals AB	\$19.00	\$55.00	223	1.8	In stock	688 Delete
COM-004	Laptop Stand Aluminum	Computing	Nordic Peripherals AB	\$12.00	\$34.00	256	2.43	In stock	688 Delete
COM-005	27" QHD IPS Monitor	Computing	Shenzhen Nova Components	\$118.00	\$249.00	185	1.2	In stock	688 Delete
COM-006	4K Capture Card	Computing	Nordic Peripherals AB	\$52.00	\$129.00	0	0	Out of stock	688 Delete
PWR-001	65W GaN Charger	Power	Pacific Power Labs	\$11.00	\$36.00	483	4.03	In stock	688 Delete
PWR-002	20,000mAh Power Bank	Power	Pacific Power Labs	\$16.00	\$49.00	382	3.37	In stock	688 Delete
PWR-003	Wireless Charging Pad 15W	Power	Pacific Power Labs	\$8.50	\$29.00	248	2.33	In stock	688 Delete
PWR-004	UPS 850VA Line-Interactive	Power	Pacific Power Labs	\$58.00	\$139.00	42	0.77	In stock	688 Delete
PWR-005	Solar Camping Charger	Power	Pacific Power Labs	\$21.00	\$59.00	0	0	Out of stock	688 Delete
SHH-001	Smart Plug (2-pack)	Smart Home	Shenzhen Nova Components	\$9.00	\$27.00	398	3.13	In stock	688 Delete
SHH-002	Wi-Fi Video Doorbell	Smart Home	Shenzhen Nova Components	\$34.00	\$95.00	182	1.57	In stock	688 Delete
SHH-003	Smart LED Bulb E27 RGBWW	Smart Home	Shenzhen Nova Components	\$5.50	\$19.00	401	4.57	In stock	688 Delete
SHH-004	Zigbee Motion Sensor	Smart Home	Shenzhen Nova Components	\$8.00	\$25.00	40	1.2	Low stock	688 Delete
SHH-005	Smart IR Remote Hub	Smart Home	Shenzhen Nova Components	\$12.00	\$35.00	88	0	In stock	688 Delete

Inventory — SKU catalog with search + category/status filters; on-hand qty is intentionally read-only

StockPilot

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Demo Operator
demo@stockpilot.app
Sign out

Stock movements

The append-only ledger — every change to on-hand stock lives here with a reason.

Record a movement

PRODUCT
ACC-001 — Adjustable Phone Tripod (199 on hand)
QUANTITY
1
REASON
Online order
REFERENCE (OPTIONAL)
SO-12345 / PO-2026-0405
Record movement

Search SKU, product, reason...

All

In

Out

Adjust

400 entries

WHEN	SKU	PRODUCT	TYPE	QTY	BALANCE	REASON
Jul 10, 7:00 PM	COM-005	27" QHD IPS Monitor	ADJUST	+1	185	Customer return/refund
Jul 10, 8:00 PM	SHH-003	Smart LED Bulb E27 RGBWW	OUT	-7	401	828 order - SO-41688
Jul 10, 8:00 PM	SHH-001	Smart Plug (2-pack)	OUT	-3	398	Retail order - SO-81402
Jul 10, 8:00 PM	AUD-002	True Wireless Earbuds Pro	OUT	-11	504	Retail order - SO-12107
Jul 10, 5:00 PM	COM-005	27" QHD IPS Monitor	OUT	-1	184	Marketplace order - SO-41690
Jul 10, 5:00 PM	CAB-002	USB-C Hub 7-in-1	OUT	-5	380	828 order - SO-28348
Jul 10, 4:00 PM	PWR-003	Wireless Charging Pad 15W	OUT	-4	248	828 order - SO-78879
Jul 10, 4:00 PM	COM-002	Mechanical Keyboard TKL	OUT	-6	402	Online order - SO-11009
Jul 10, 4:00 PM	COM-002	Wireless Ergonomic Mouse	OUT	-7	545	Online order - SO-70017
Jul 10, 3:00 PM	ACC-003	Cable Management Kit	OUT	-4	274	Marketplace order - SO-76742
Jul 10, 3:00 PM	ACC-002	Laptop Sleeve 14" Felt	OUT	-2	182	Marketplace order - SO-13386
Jul 10, 3:00 PM	PWR-004	UPS 850VA Line-Interactive	OUT	-1	42	Retail order - SO-46768
Jul 10, 3:00 PM	AUD-004	Bluetooth Party Speaker XL	OUT	-1	34	Retail order - SO-78959
Jul 10, 1:00 PM	PWR-003	15,000mAh Power Bank	OUT	-6	182	828 order - SO-78100

Movements — the append-only ledger with a record form for IN / OUT / signed ADJUST

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Purchase orders

Draft — ordered — received. Receiving writes IN movements and updates stock in one transaction.

New purchase order

PO-2026-0404 IN DRAFT
Rhine Cable GmbH - ordered Jul 9, 2026
SKU PRODUCT QTY UNIT COST LINE TOTAL
CAB-005 Thunderbolt 4 Cable 1m 110 \$15.00 \$1,650.00
PWR-005 Solar Camping Charger 90 \$21.00 \$1,890.00
Total \$3,540.00

PO-2026-0403 ORDERED
Nordic Peripherals AB - created Jul 4, 2026 - ordered Jul 5, 2026 - expected in 3d
SKU PRODUCT QTY UNIT COST LINE TOTAL
COM-004 4K Capture Card 130 \$52.00 \$6,760.00
Total \$6,760.00

PO-2026-0402 RECEIVED
Pacific Power Labs - created Jun 17, 2026 - ordered Jun 18, 2026 - received Jul 5, 2026
SKU PRODUCT QTY UNIT COST LINE TOTAL
PWR-001 65W GaN Charger 320 \$11.00 \$3,520.00
PWR-002 20,000mAh Power Bank 280 \$16.00 \$4,480.00
Total \$8,000.00

PO-2026-0401 RECEIVED
Rhine Cable GmbH - ordered Jun 4, 2026 - received Jun 6, 2026 - received Jun 14, 2026
SKU PRODUCT QTY UNIT COST LINE TOTAL
CAB-001 USB-C to HDMI Cable 2m 500 \$4.20 \$2,100.00
CAB-002 USB-C Hub 7-in-1 260 \$13.00 \$3,380.00
CAB-003 Braided Lightning Cable 1m 450 \$2.80 \$1,260.00
Total \$6,740.00

Purchase orders — DRAFT → ORDERED → RECEIVED status machine, with a "prefill low-stock SKUs" builder

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Suppliers

Lead times drive the advisor's urgency map — a 21-day supplier needs three weeks of warning.

SUPPLIER	CONTACT	COUNTRY	LEAD TIME	SKUS	LOW-STOCK SKUS
Baltic Electronics OÜ	sales@balticelec.eu	Estonia	9 days	4	0
Kaizen Audio Works	info@kaizenaudio.jp	Japan	14 days	5	0
Nordic Peripherals AB	orders@nordicper.se	Sweden	10 days	5	1
Pacific Power Labs	info@pplabs@pacificpower.tw	Taiwan	18 days	5	1
Rhine Cable GmbH	vertrieb@rhinecable.de	Germany	7 days	5	1
Shenzhen Nova Components	orders@novacom.cn	China	21 days	6	1

Suppliers — directory with lead times and per-supplier SKU / low-stock counts

The mechanics that make this interview-worthy:

- Ledger, not a number.** On-hand quantity is read-only in the UI — it is the running balance of the immutable **StockMovement** ledger. Adding a product writes an "Opening balance" IN movement in the same transaction.
- Row-locked writes.** Recording a movement locks the product row (**SELECT ... FOR UPDATE**) inside a transaction; an OUT that would oversell is rejected with **HTTP 409**, and every movement snapshots the resulting balance.

- **Atomic receiving.** Receiving a PO increments stock *and* writes IN movements in one Prisma transaction. Deleting a SKU is blocked while it sits on a DRAFT/ORDERED PO.
- **Velocity-driven.** 30-day sales velocity, days-of-cover, dead-stock (no sales ≥ 45 days), and netting against inbound open-PO quantity drive both the alerts and the advisor.

4 • The AI Reorder Advisor

The headline feature: turn the raw ledger into a ranked replenishment plan — what to order this week, and what is dead stock — with a written rationale per SKU.

StockPilot

- Dashboard
- Inventory
- Movements
- Purchase orders
- Suppliers
- Reorder advisor**

Reorder advisor

30-day velocity × supplier lead time × on-hand cover, ranked by what bites first.

Plan summary CLAUDE PLANNER · JUL 14, 2026, 12:35 PM

We have analyzed the inventory status for StockPilot. Two items (AUD-004 and SMH-004) are nearing or have reached their reorder points and require replenishment soon to cover their lead times. Two items (AUD-005 and SMH-005) show no historical demand despite having stock on hand, making them dead stock. We recommend 0 replenishment and a clearance campaign for these. All other SKUs currently maintain healthy stock levels.

Order this week

Cover runs out shortly after the lead time

Item	Status	Details	Action
AUD-004 Bluetooth Party Speaker XL	Order soon	On-hand stock is at 34 units, which is close to the reorder point of 30, offering 32 days of cover. Ordering this week ensures we do not stock out given the 14-day lead time. Suggested quantity is set to the default order size of 80.	Suggest ordering 80 units
SMH-004 Zigbee Motion Sensor	Order soon	On-hand stock of 40 units has reached the reorder point of 40, providing 33 days of cover. A new order is needed this week to prevent stockouts during the 21-day lead time. We suggest the default order quantity of 100.	Suggest ordering 100 units

Dead stock

No demand — cash tied up on the shelf

Item	Status	Details	Action
AUD-005 Vinyl Turntable Classic	Dead stock	This product has never recorded a sale and has 64 units sitting on hand. No further orders should be placed; a clearance strategy is advised to liquidate this dead stock.	Do not reorder
SMH-005 Smart IR Remote Hub	Dead stock	No sales have ever been recorded for this item, and we are holding 88 units in stock. We recommend 0 replenishment and running a markdown/clearance promotion to free up space.	Do not reorder

Ready to act? [Create a purchase order](#) →

Demo Operator
demo@stockpilot.app
Sign out

Reorder advisor — a generated plan with an executive summary, an "Order this week" group (suggested quantities + days-of-cover) and a "Dead stock" group (clearance advised). The badge shows the AI planner produced it.

How the AI works — a three-tier fallback chain (in `src/lib/advisor.ts` → `generateAndSavePlan`):

- **1 • Claude** — the Anthropic SDK with **forced tool-use** (`report_reorder_plan` , a JSON input-schema), so the model must return structured suggestions — no fragile JSON parsing. Model `claude-sonnet-5` (override `ADVISOR_MODEL`).
- **2 • Gemini** — a raw REST call to Google's `gemini-flash-latest` using a `responseSchema` for structured JSON. A free-tier fallback with no extra dependency.
- **3 • Heuristic** — a transparent rule engine (cover lead time + a 30-day buffer, flag dead stock, net against inbound). Always works with no API key.

Both LLM tiers receive a **byte-for-byte identical prompt** built by `advisorPrompt()` from a per-SKU signal table, and each run is tagged `AI` or `HEURISTIC` and persisted (`AdvisorRun` → `ReorderSuggestion`).

Each suggestion carries an urgency (CRITICAL / SOON / OK / DEAD), a suggested order quantity, days-of-cover, and a rationale — so the plan is explainable, not a black box.

5 • Likely interview questions

Q: Why derive on-hand from a ledger instead of a quantity column?

Auditability and correctness. Every change is an immutable, timestamped movement with a balance snapshot, so you can reconstruct stock at any point in time and never lose history to an overwrite.

Q: How do you prevent overselling under concurrent writes?

The movement handler opens a transaction, locks the product row with `SELECT ... FOR UPDATE`, re-checks available stock, and rejects an oversell with a 409 — so two simultaneous OUT movements can't both succeed past zero.

Q: Is the AI making up the numbers?

No — velocity, days-of-cover and dead-stock flags are computed deterministically. The LLM only writes the ranked plan and rationale, and if no key is set the heuristic planner produces the same shape, so the app always works.

Q: How is structured output guaranteed from the LLM?

Tool-use: the output shape is a tool's input schema and `tool_choice` forces the model to call it (Claude), or a `responseSchema` (Gemini). I read typed arguments, never parse free text.